1 Data Overview and Initial Exploration

- **Dataset Source:** Kaggle House Prices: Advanced Regression Techniques
- **Initial Dimensions:** 1460 training samples (rows), 81 feature columns.
- **Data Type Distribution:** Includes 38 numerical features (`int64`, `float64`) and 43 categorical/text features (`object`).
- **Key Observations:** The dataset contains a significant amount of missing values. Furthermore, the large number of initial features makes direct application to linear models prone to overfitting and the curse of dimensionality.

2 Feature Engineering and Missing Value Imputation

To maximize data value and mitigate noise, we adopted a stratified treatment strategy for missing values rather than blind deletion:

2.1 Extracting Informative Missingness

- **Strategy:** Features with extremely high missing rates ($> 80\%$, e.g., `PoolQC`, `Alley`, `Fence`, `MiscFeature`) were transformed into binary features (1: Present / 0: Absent), creating new columns such as `HasPool`.
- **Decision Rationale:** This approach retains highly valuable signals for house price prediction (e.g., a luxury home possessing a pool) while resolving model instability caused by overly sparse original columns. The original high-missing-rate columns were subsequently dropped.

2.2 Numerical Feature Imputation

- **Strategy:** Implemented an automated imputation mechanism based on **Skewness**. The absolute skewness of each numerical feature was evaluated:
  - **Highly Skewed ($|\textbf{Skewness}| > 1.0$):** Indicates the presence of extreme outliers (e.g., ultra-luxury mansions). Imputed using the **median**.
  - **Roughly Symmetric ($|\textbf{Skewness}| \leq 1.0$):** Imputed using the **mean**.
- **Decision Rationale:** This dynamic strategy adapts to the underlying distribution of each individual feature, preserving the original statistical properties much more accurately than a uniform median imputation.

2.3 Categorical Feature Imputation

- **Strategy:** Abandoned uniform mode imputation in favor of a dual-track decision mechanism combining **statistical distribution** and **domain knowledge**:
  - **Extremely Low Missing Rate with Dominant Class (e.g., `Electrical`):** With a missing rate of only 0.1% and a mode dominance of 91.4%, this indicates random human omission. Imputed using the **mode**.
  - **Domain-Specific Correlated Missingness (e.g., `Garage` and `Bsmt` families):** Observed uniform missing rates across related features (5.5% for garage, 2.5% for basement). Domain knowledge dictates this represents the physical absence of the facility (i.e., "No Garage"), not missing data. Forced mode imputation would artificially synthesize non-existent amenities, thus these were imputed with **"None"**.
  - **High Missing Rate Features (e.g., `MasVnrType`, `FireplaceQu`):** Missing rates between 40%-60% indicate the general absence of these specific upgrades. Imputed with **"None"**.
- **Decision Rationale:** This prevents distribution distortion caused by blind mode imputation and ensures the engineered features strictly adhere to real-world real estate logic, thereby increasing the reliability of the predictive model.

3 Categorical Encoding

Since machine learning algorithms (Linear, Ridge, Lasso, SVR) cannot directly process text, strings must be converted into numerical representations:

- **Strategy:** Applied **One-Hot Encoding** utilizing `pd.get_dummies(drop_first=True)`.
- **Dimensionality Change:** After expanding over 40 categorical features, the total number of features expanded from 77 to 241.
- **Decision Rationale:** Using `drop_first=True` drops the first category level to bypass the **Dummy Variable Trap**. This ensures the stability of subsequent linear regression models and mitigates multicollinearity issues.

4 Target Variable and Feature Splitting

- **Feature Matrix (X) and Target Variable (y):** `SalePrice` was isolated as `y`, while the remaining 240 features constituted the feature matrix `X`.
- **Train/Test Split:** Utilized `scikit-learn` to split the dataset at an 80:20 ratio (Train: 80%, Test: 20%). A fixed `random_state` was configured to guarantee consistency and reproducibility in future model comparison benchmarks.

4.1 Baseline Linear Regression

4.2 Baseline Linear Regression (OLS)

To establish a rigorous performance benchmark for evaluating subsequent regularized models and advanced ensemble algorithms, this study first implemented a baseline model utilizing **Ordinary Least Squares (OLS)** linear regression.

4.2.1 Target Variable Transformation

Initial exploratory data analysis revealed that the target variable, `SalePrice`, exhibited a significant right-skewed distribution, as illustrated in the left panel of **Figure 1**. Such skewness can lead linear models to be overly sensitive to high-value outliers, potentially compromising generalizability.

To address this, a log-transformation was applied: $y = \log(1 + \text{SalePrice})$. As shown in the right panel of **Figure 1**, the transformed data approximates a normal distribution (Gaussian-like). This transformation satisfies the homoscedasticity and normality assumptions of linear regression and converts absolute price errors into relative percentage errors, ensuring more robust weight estimation across different price tiers.

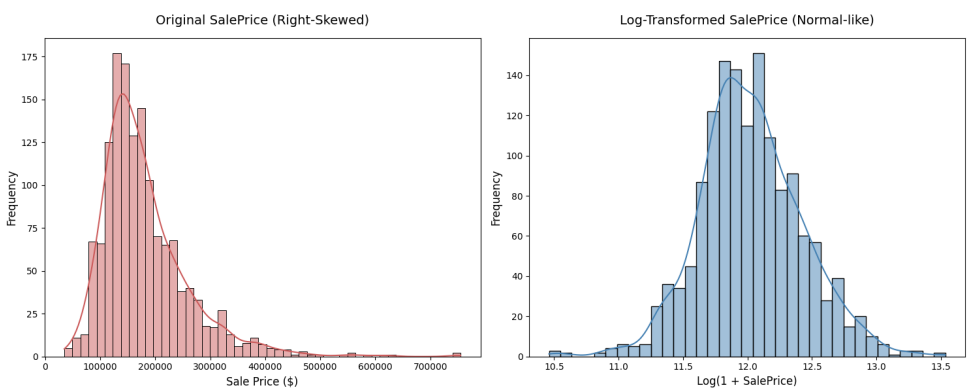


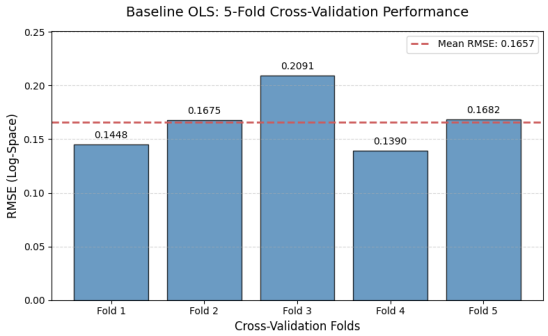
Figure 1: Distribution of SalePrice: Original skewed data (left) vs. Log-transformed normal-like data (right).

4.2.2 Model Evaluation and Visual Analysis

The baseline model was evaluated using **5-Fold Cross-Validation** on the development set (80% of the data) with a shuffle mechanism (`random_state=42`) to ensure stability.

Baseline Model Diagnostics:
To comprehensively evaluate the performance of the Baseline OLS model, we analyzed its predictive distribution and stability through two complementary visualizations, as shown in Figure 2.

(a) Actual vs. Predicted Plot



(b) 5-Fold Cross-Validation Scores

Figure 2: Diagnostic visualizations of the Baseline OLS model, illustrating high variance and weight instability.

Based on the combined evidence from Figure 2, we identify the following challenges:

- **Catastrophic Underprediction and Weight Instability:** As illustrated in Figure 2a, the model suffers from extreme predictive failures. Specifically, a standard property with an **actual log value of $\sim$12.5** was severely underpredicted at a **log value of $\sim$9.75**. This magnitude proves that the OLS model overreacts to specific feature combinations in the 240-dimensional space, leading to erratic coefficient estimates.
- **Performance Fluctuation in CV:** Figure 2b confirms this instability, with RMSE scores fluctuating violently between Fold 3 (0.2091) and Fold 4 (0.1390). This sensitivity to data splits is a hallmark of **High Variance** and **Overfitting**.

These findings establish a definitive necessity for implementing **Ridge and Lasso Regression** to constrain the coefficient magnitudes and restore model robustness.

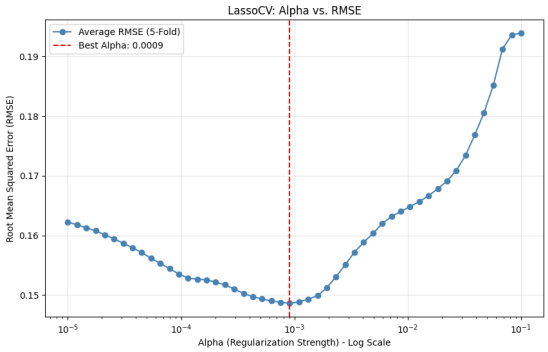
Table 1: Performance Summary of Baseline Linear Regression

Metric	**RMSE (Log-Space)**
5-Fold CV Average RMSE	0.1657
Final Test RMSE	0.2048

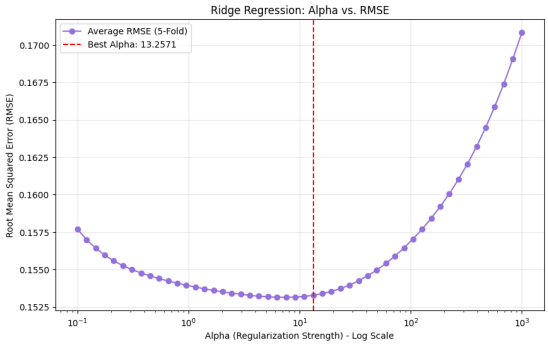
4.3 L1 and L2 Regularization (Lasso and Ridge)

4.4 Hyperparameter Tuning: Alpha Path Analysis

To determine the optimal regularization strength, we performed a log-scale grid search using 5-Fold Cross-Validation for both Lasso and Ridge models.



(a) LassoCV: Optimal $\alpha = 0.0009$



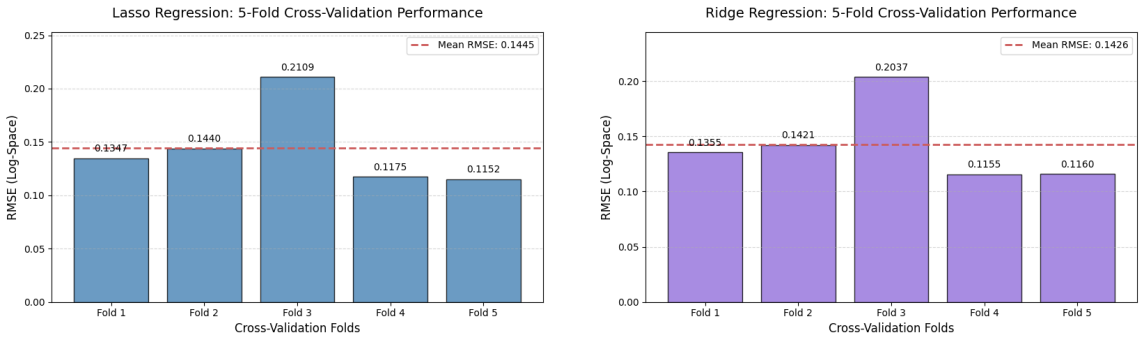
(b) RidgeCV: Optimal $\alpha \approx 13.25$

Figure 3: Hyperparameter tuning paths. The red dashed lines indicate the α values that minimize the mean Cross-Validation RMSE.

- **Under-regularization (Left side):** At extremely low α values, the penalty is negligible, and the models behave similarly to the unconstrained Baseline OLS. They suffer from high variance by capturing too much dataset noise, leading to elevated cross-validation errors.
- **Over-regularization (Right side):** Conversely, as α increases excessively, the models face severe penalties. Critical feature weights are shrunk to zero (Lasso) or near-zero (Ridge), resulting in high bias and underfitting, which causes the RMSE to spike drastically.
- **The Optimal Sweet Spot (Red dashed line):** The global minimum on each curve represents the optimal balance between bias and variance. Notably, Lasso achieves its optimum at a highly sensitive, small scale ($\alpha = 0.0009$), whereas Ridge optimizes at a vastly larger scale ($\alpha \approx 13.25$). This discrepancy is driven by their mathematical formulations: the L1 norm (absolute sum) in Lasso applies a much harsher shrinkage per unit of α compared to the L2 norm (squared sum) in Ridge.

By systematically selecting these specific α parameters, we ensure the models effectively filter out idiosyncratic noise while retaining robust predictive signals.

4.5 Model Stabilization: Cross-Validation Performance



(a) Lasso CV (Mean RMSE: 0.1445)

(b) Ridge CV (Mean RMSE: 0.1426)

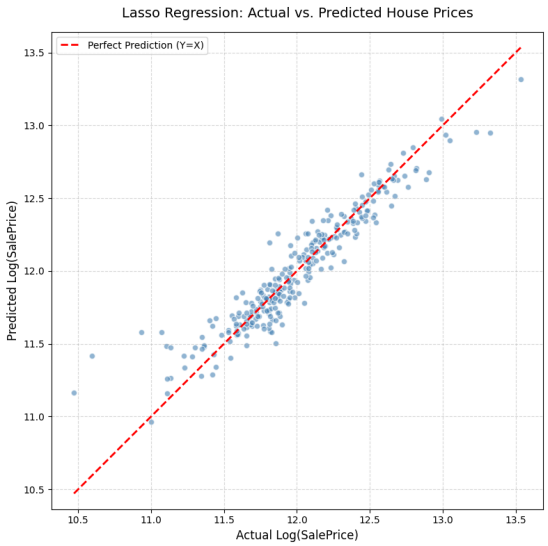
Figure 4: Cross-Validation performance of regularized models.

- As shown in **Figure 4**, both models achieved lower overall mean RMSEs (Lasso: 0.1445, Ridge: 0.1426) compared to the unregularized baseline, successfully dampening the extreme predictive fluctuations across most data splits.
- However, a critical observation must be made regarding the persistent instability in **Fold 3**. Despite the application of optimal L1 and L2 penalty terms, Fold 3 exhibits a disproportionately high error (Lasso: 0.2109, Ridge: 0.2037) relative to the other folds, which tightly cluster between 0.11 and 0.14.
- This prominent residual variance reveals a fundamental limitation: while regularization effectively constrains coefficient magnitudes to prevent overfitting, it cannot fully mitigate variance if the underlying data slice contains severe outliers or structural anomalies. The resilient spike in Fold 3 suggests the presence of complex, non-linear relationships within that specific validation subset that fundamentally violate the assumptions of linear models, warranting future exploration with robust, non-linear algorithms (e.g., Tree-based models).

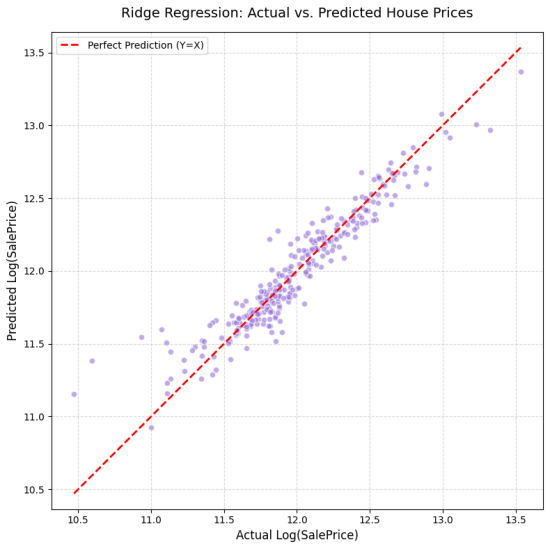
4.6 Outlier Mitigation in Individual Predictions

While the cross-validation metrics reveal the macro-level behavior of the regularized models across different data splits, analyzing the Actual vs. Predicted scatter plots provides crucial micro-level insights into individual property valuations.

This visual diagnostic reveals the true triumph of regularization: the elimination of catastrophic weight instability.



(a) Lasso Regression



(b) Ridge Regression

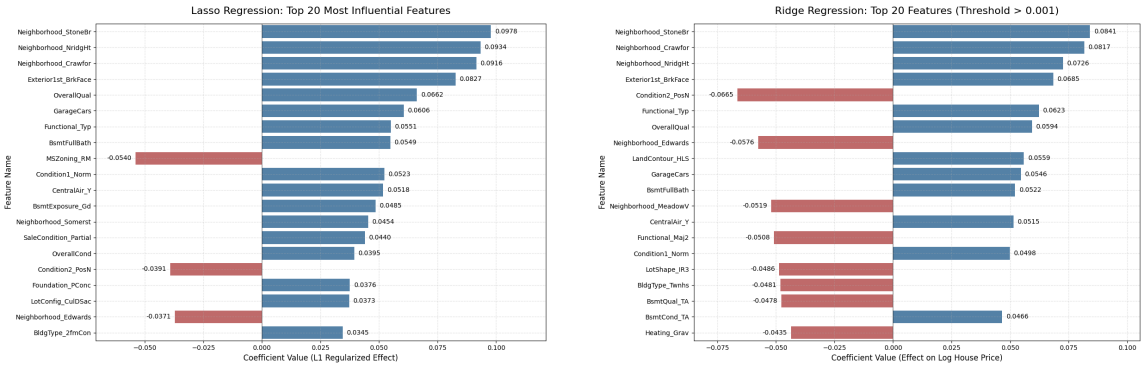
Figure 5: Actual vs. Predicted distributions for regularized models. Both techniques successfully eliminate the severe underprediction outliers observed in the baseline OLS.

As illustrated in **Figure 5**, both Lasso and Ridge models successfully bounded the predictions closely to the 45-degree reference line. Most notably, the catastrophic underprediction present in the Baseline OLS model—where a standard property (Actual Log $\sim$12.5) was severely underestimated to a Log value of $\sim$9.75—has been completely eradicated.

By penalizing large coefficients, both L1 and L2 regularizations acted as a crucial safety net. Even if they could not perfectly flatten the variance across all data folds (as seen in Section 4.5), they effectively prevented the model from overreacting to idiosyncratic noise, ensuring that no individual property suffers from extreme valuation errors.

4.7 Feature Insights: Sparsity vs. Shrinkage

While both models improved predictive stability, their handling of the 240-dimensional feature space differed fundamentally due to their distinct penalty terms.



(a) Lasso: Feature Selection (Sparsity)

(b) Ridge: Coefficient Shrinkage

Figure 6: Top 20 most influential features.

Model	Penalty	Total Features	Retained	Suppressed	Criteria
Lasso	L1	250	**92**	158	Coef = 0
Ridge	L2	250	**217**	33	$\lvert\text{Coef}\rvert < 0.001$

Table 2: Quantitative comparison of feature selection and shrinkage effects between Lasso and Ridge regression.

4.7.1 Quantitative Impact: Sparsity vs. Shrinkage

As quantified in **Table 2**, the fundamental difference between L1 and L2 regularization is explicitly revealed in their final active feature counts. Lasso effectively acts as a strict automated feature selector; it aggressively forces the coefficients of 158 less informative or highly collinear features to exactly zero, retaining a sparse subset of only 92 critical predictors. In contrast, Ridge employs continuous coefficient shrinkage. By retaining 217 features with absolute weights greater than 0.001, Ridge demonstrates its mathematical preference to distribute the penalty evenly across correlated variables rather than eliminating them entirely.

4.7.2 Qualitative Consensus on Market Drivers

Despite these opposing mathematical mechanics, **Figure 6** demonstrates that both models reach a striking consensus on the most impactful drivers of property value.

- **Top Value Additions:** Premium neighborhood indicators, such as `Neighborhood_StoneBr` and `Neighborhood_NridgHt`, along with visual and structural quality metrics like `OverallQual`, consistently dominate the top positive coefficients.

- **Top Value Detractions:** Similarly, adverse conditions like `Condition2_PosN` or specific lower-valued areas such as `Neighborhood_Edwards` receive the heaviest negative penalties.

Ultimately, this dual analysis highlights the strategic trade-off in regularization: Lasso provides a highly interpretable, streamlined model by isolating the absolute strongest signals, whereas Ridge preserves the nuanced, collective predictive power of the entire feature space while safely dampening its overall variance.

5 Advanced Ensemble Modeling: Gradient Boosting

5.1 Hyperparameter Tuning Strategy

5.2 Validation Curve Analysis

6 Feature Importance and Model Comparison

6.1 The Dominance of Quality and Area

6.2 Linear vs. Non-Linear Discrepancy (Masking Effect)

7 Conclusion